

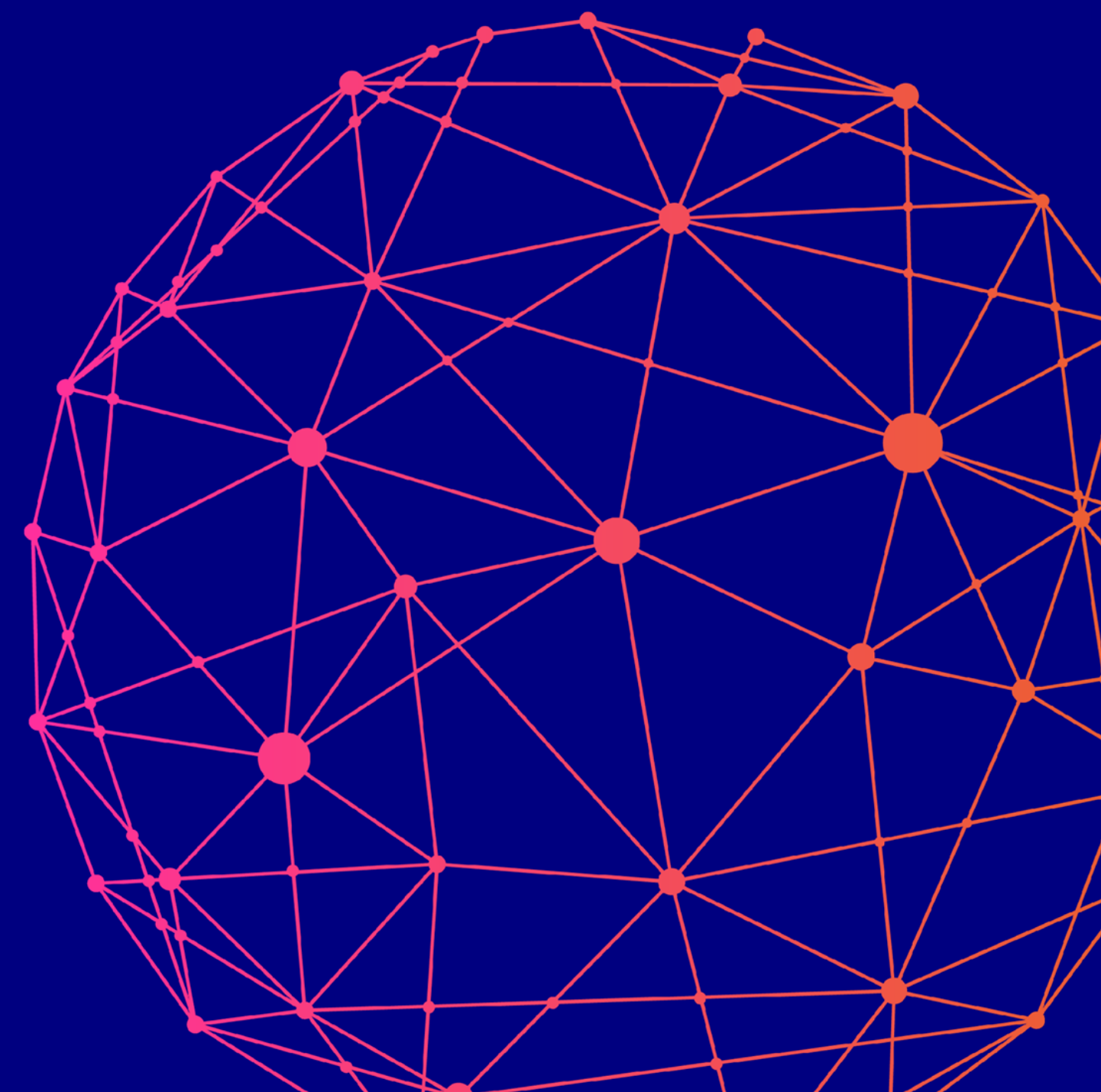
Data Spaces Symposium

Accelerating adoption. Increasing impact.

Mondragon-X: An Interoperable
Industrial Data Space powered
by Open Standards

Iñaki Paz ipaz@lksnext.com
Innovation Project Manager – DataSpace Lead
LKS Next

This project has been co-financed by Spanish Ministry of Digital Transformation and the Civil Service under European Union's Recovery and Resilience Facility, through indirect UNICO IPCEI program and with exp num **TSI-063400-2024-001**.



Data Spaces Symposium

Mondragon & LKS: Who we are

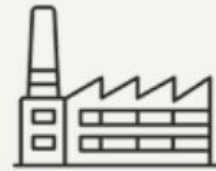


<https://www.mondragon-corporation.com/en/>

82 cooperative companies



FINANCE



INDUSTRY



RETAIL



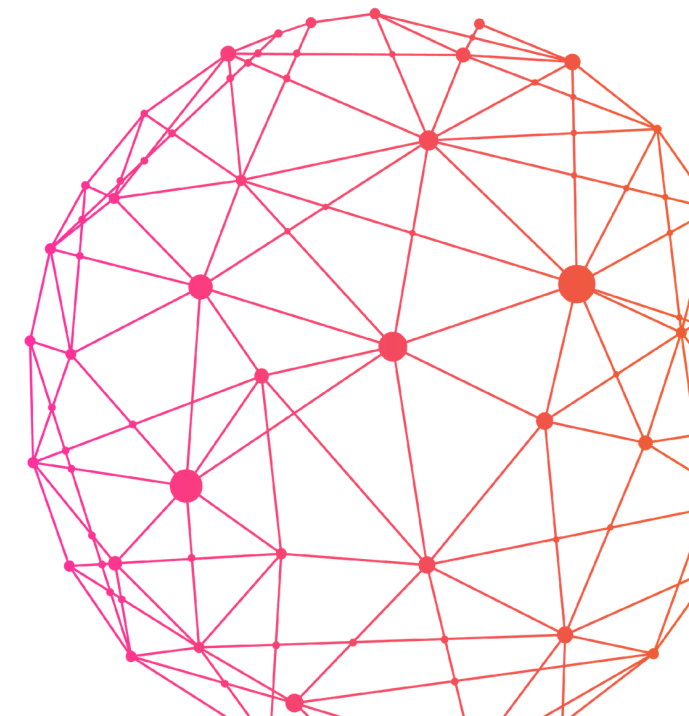
KNOWLEDGE



70.000
PEOPLE



SALES IN MORE THAN
150
COUNTRIES



Data Spaces Symposium



INDUSTRY



Mondragon & LKS: Who we are

MONDRAGON Automotive



cikautxo

BATZ

fagorederlandgroup

TAFALLA IRON FOUNDRY

MAPSA

@cenarro

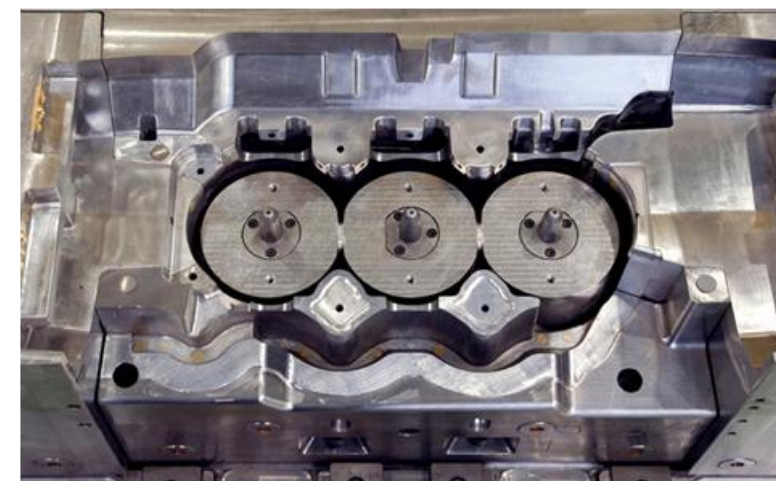
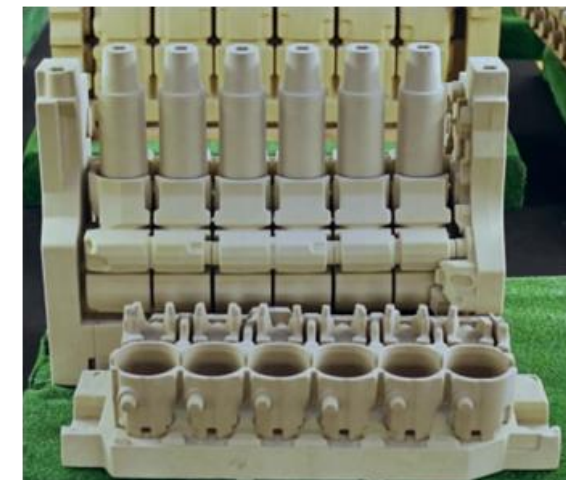
MATRICI

AK aurrenak

Loramendi

MB SISTEMAS

MAIER



Data Spaces Symposium

Mondragon & LKS: Who we are

MONDRAGON Capital Goods / Components



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Mondragon & LKS: Who we are

The logo for LKS Next, featuring the letters 'LKS' in a bold, orange, sans-serif font, with the word 'Next' in a smaller, orange, sans-serif font directly below it. The entire logo is enclosed within a thin orange square border.

LKS
Next

- We are the **biggest professional services group with Basque capital.**
- LKS Next is a cooperative company that belongs to **Mondragón.**
- **With more than 33 years of history** as an independent consultant company.
- More than 720 professionals in **multiple disciplines.**
- **Comprehensive Industrial Service** orientation
- Continued presence in the **public sector and large business groups.**

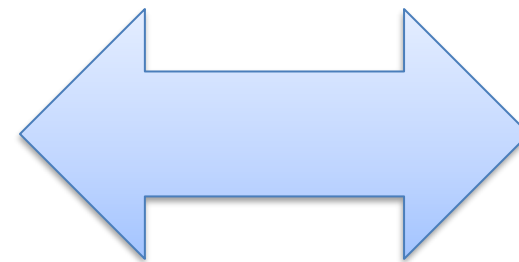
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Mondragon-X Data Space

- Dataspace for Mondragon Cooperatives
- Multi-sectorial Corporation dataspace for 82 cooperatives
- Mondragon-X must be integrated with other sectorial dataspaces to enable value chain communications!



LABORAL kutxa
GSR GESTIÓN SERVICIOS RESIDENCIALES
EROSKI
GOIMEK
Mondragon-X



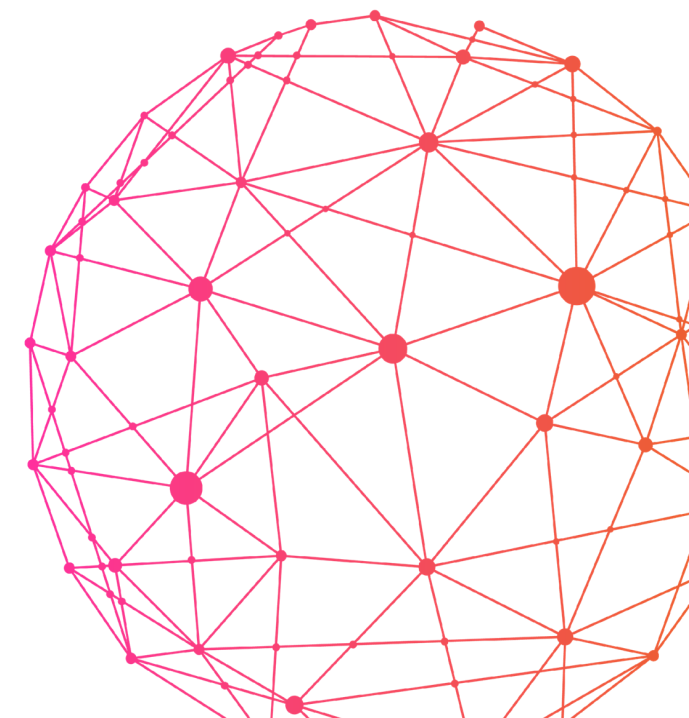
FINDIRECT.space

European Health
Data Space



Catena-X
Your Automotive Network

- How can we enable this **interoperability**?



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Interoperability Levels

Legal

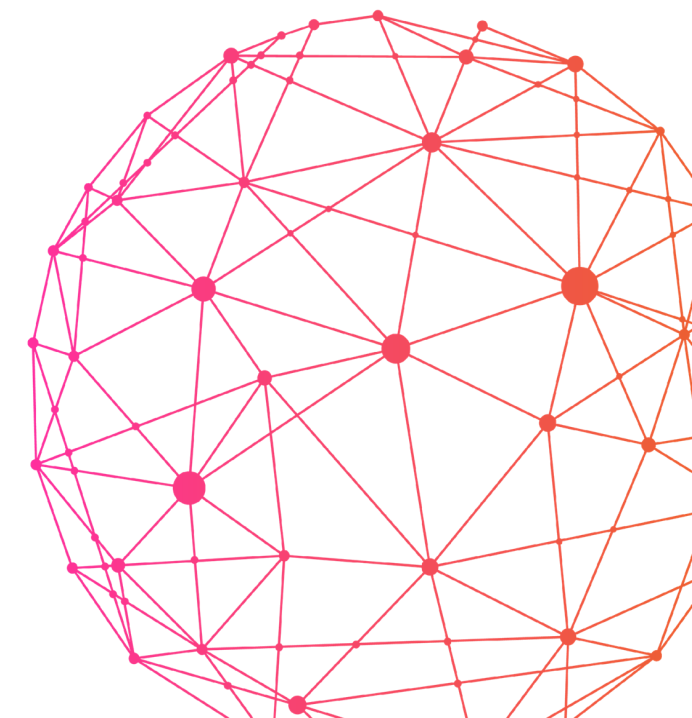
Organizational

Semantic

Technical

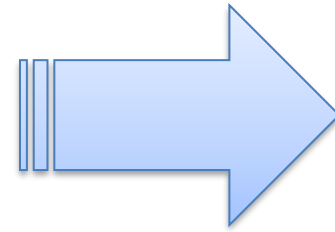
Paperwork: policy, model
standards and agreements

How could
Data Spaces be
connected?



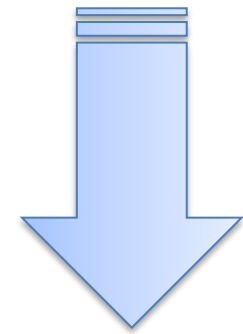
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Standards Flow into Data Space Infrastructures

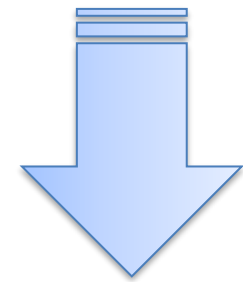
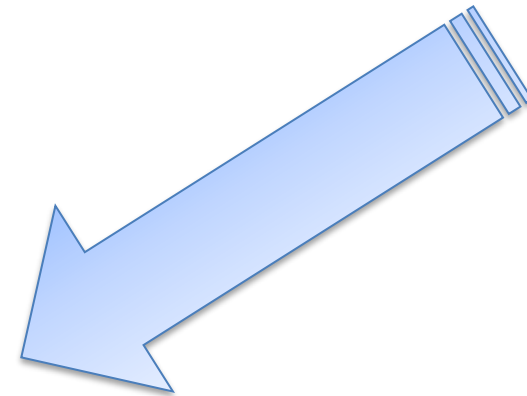


- DSP
- DCP
- DRP
- ...

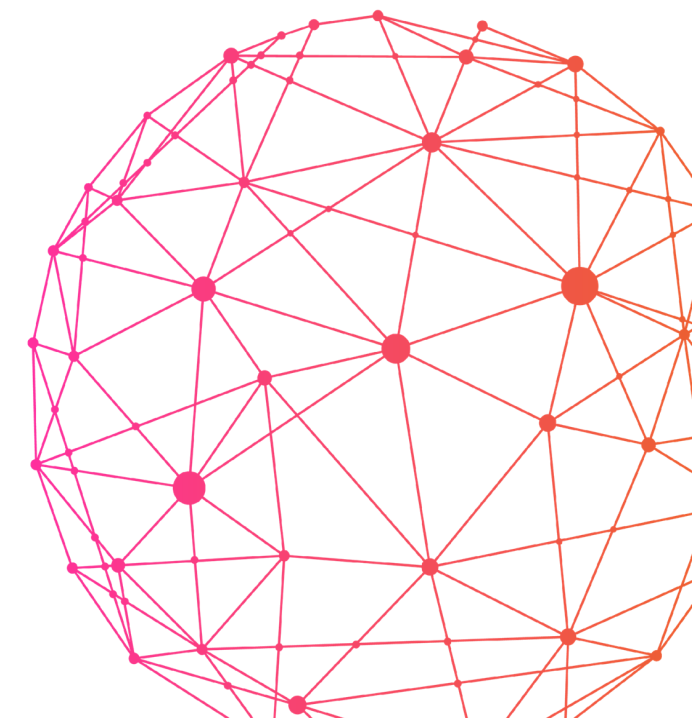
Data Space Standards



Standards Reference
Implementations (EDC)

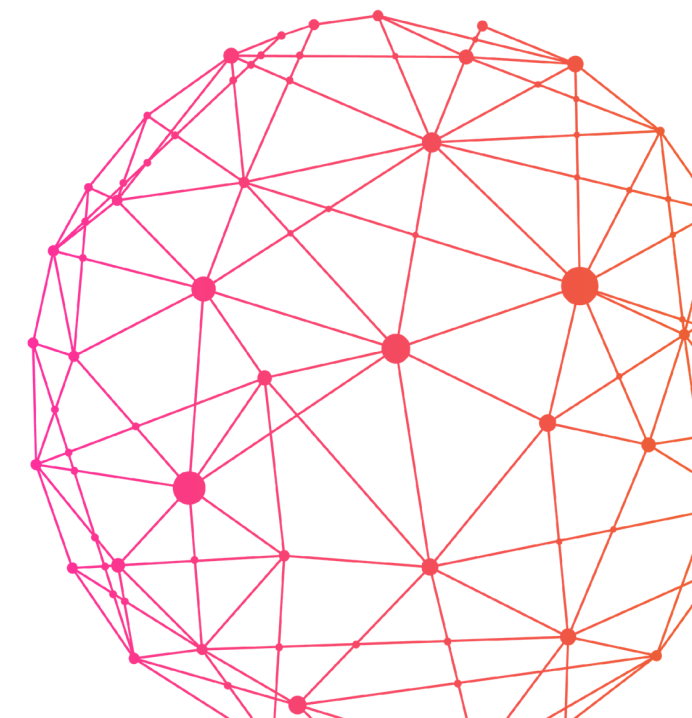
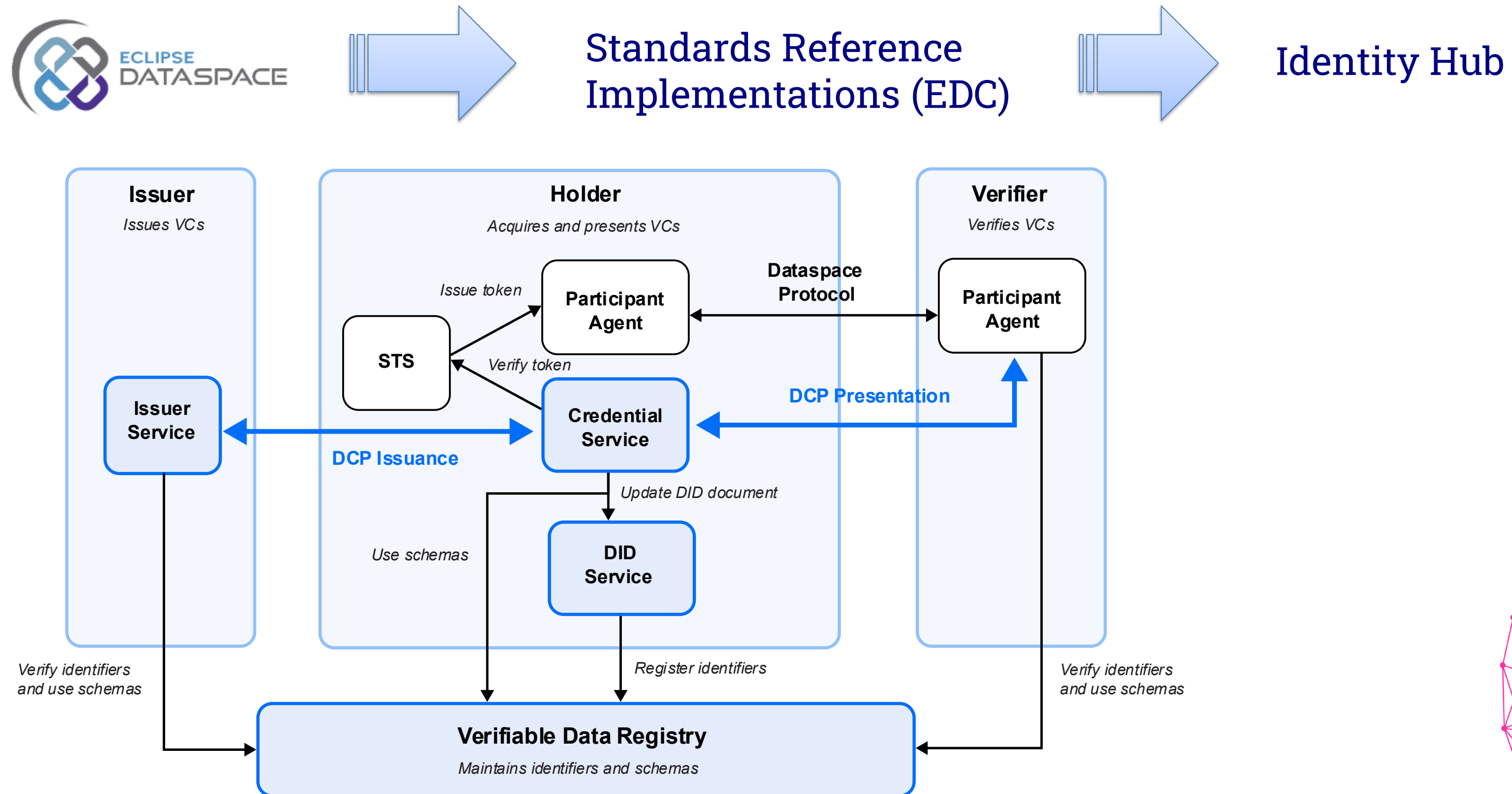


Data Space Infrastructure



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Decentralized Claims Protocol (DCP)



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Decentralized Claims Protocol (DCP)

DS Participant
Connector

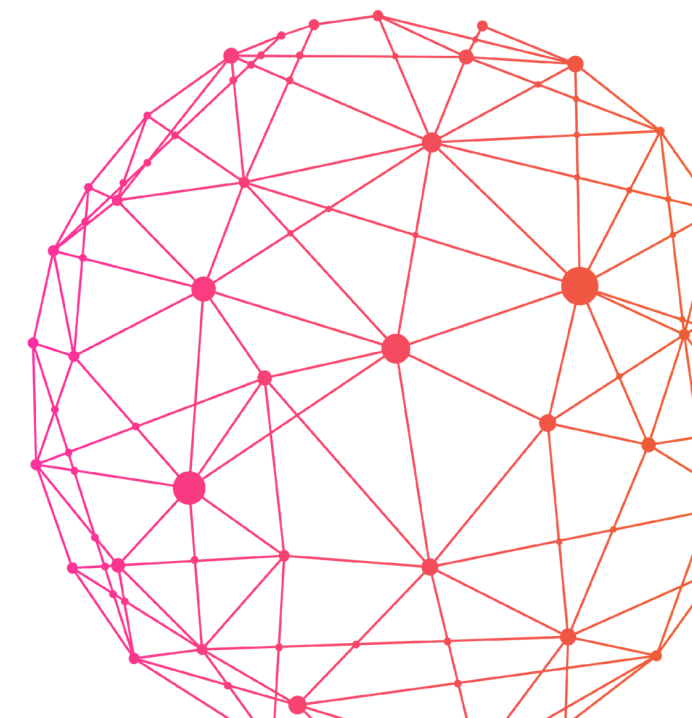
Holder / Wallet

DS Governance
Infrastructure

Issuer

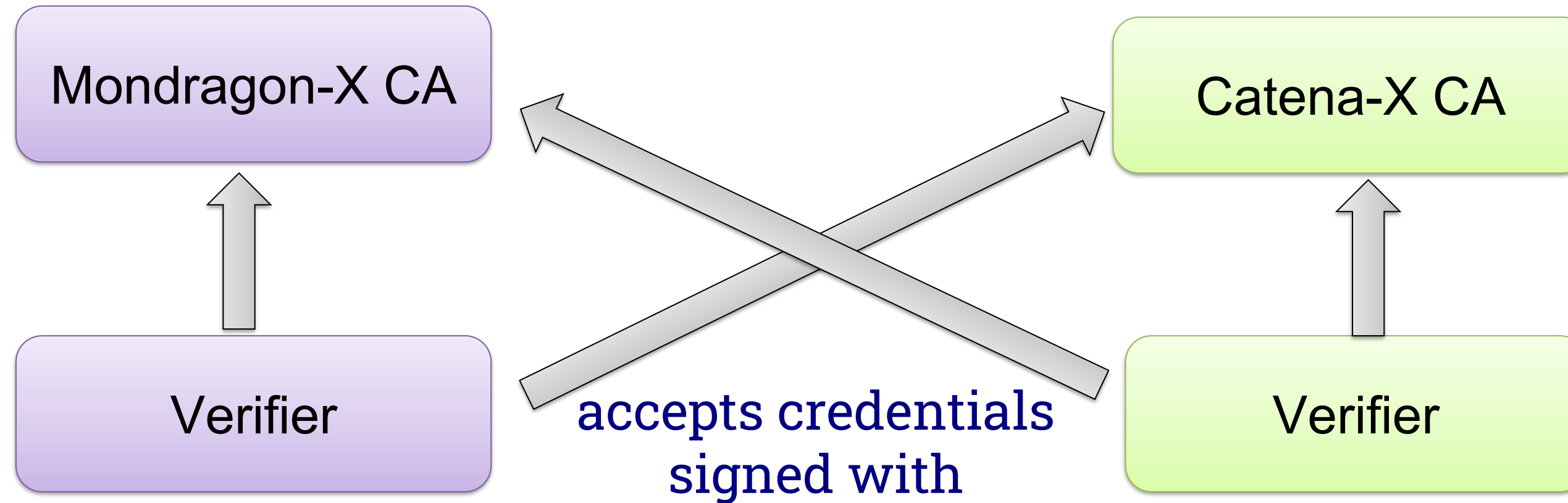
Verifier

- Issuer ISSUES Credentials signing with DS Certificates
- Verifier VERIFIES Credentials with ACCEPTED Signing Chains

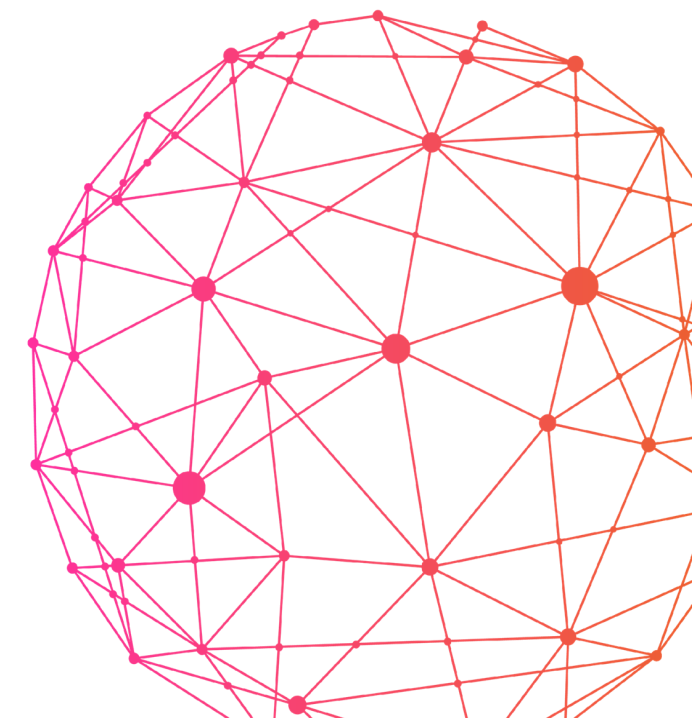


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Enabling Interoperability between Dataspaces

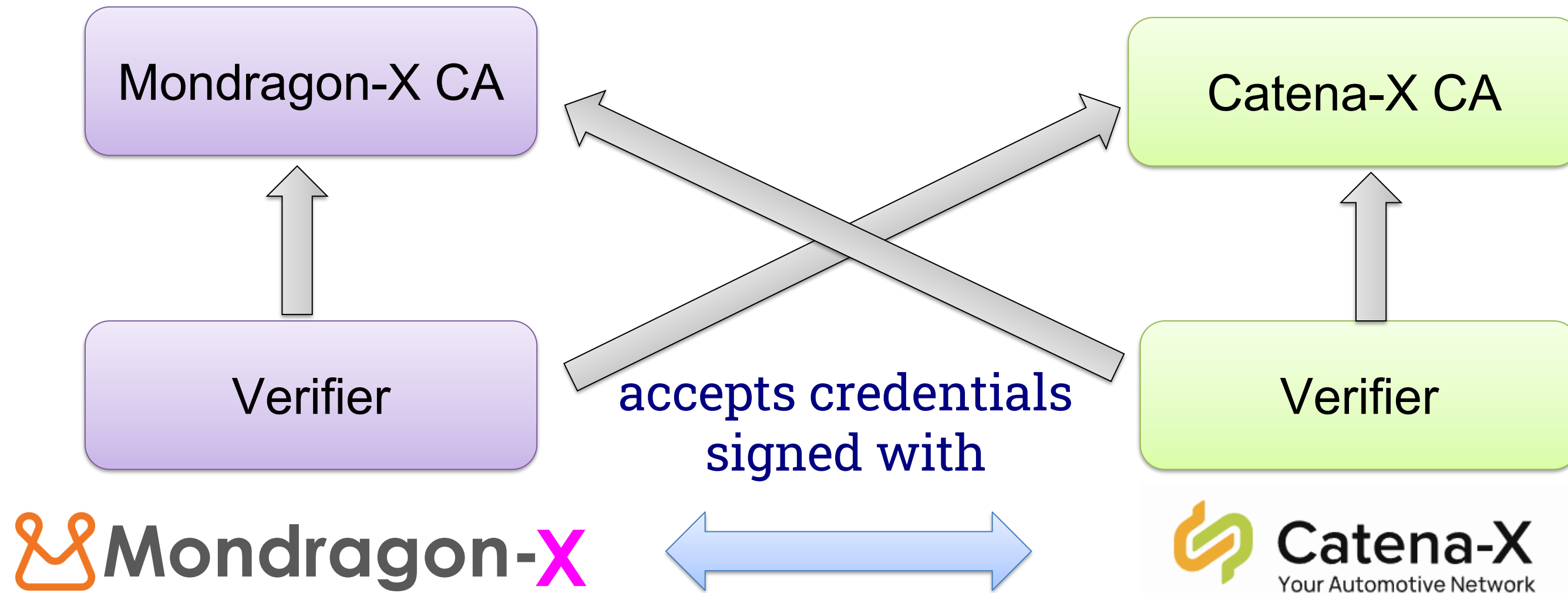


Use PKI's / DID's
Chains of Trust
and Authorities



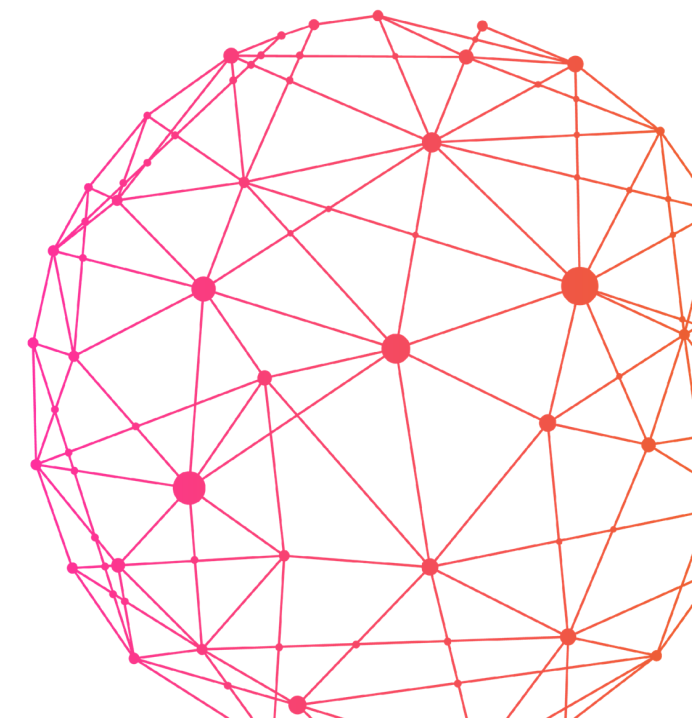
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Enabling Interoperability between Dataspaces



Use PKI's / DID's
Chains of Trust
and Authorities

- Requires Agreement on Legal, Organizational, Business and Semantics between data spaces
- Requires Sharing of Participants between data spaces (for discovery services)



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Other Scenarios



Local Sectorial Initiatives integrated in international networks, or international integration efforts (EEUU, China, etc.)



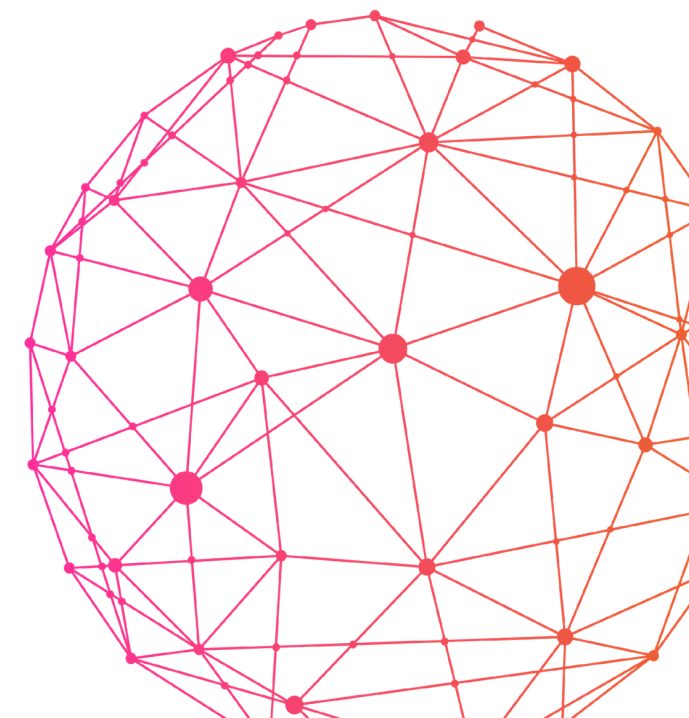
Integration of multiple small initiatives.

Just in Spain (<https://cred.digital.gob.es/kit-de-espacios-de-datos/espacios-de-datos-elegibles>):

40+ Health | 30+ Agro | 30+ Mobility | 20+ Industry



ESAUTO-X and ZAINDATA have been co-financed by the *Spanish Ministry of Digital Transformation and the Civil Service* under European Union's Recovery and Resilience Facility, through SEDIA program.



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Identity Hub Reference Implementation



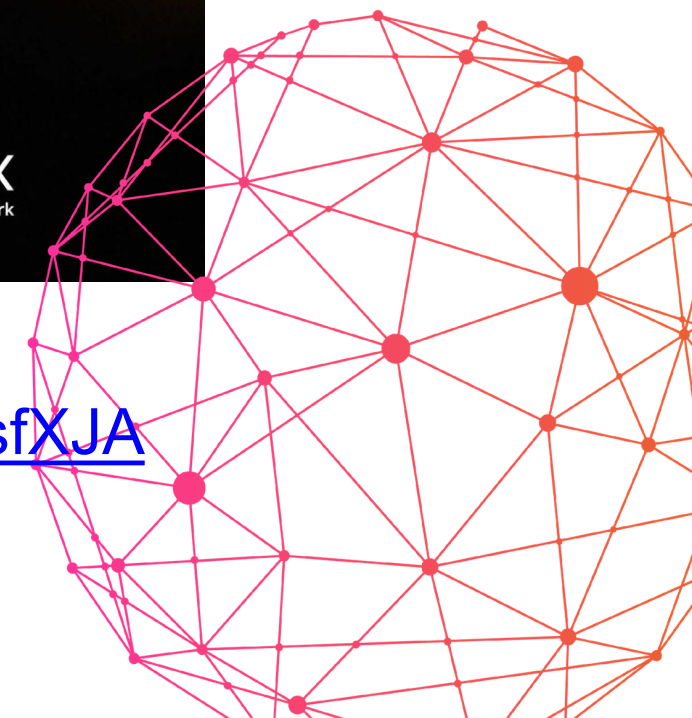
Reference Implementation Status:

- Holder / Wallet: Implemented Version supporting DID
- Issuer: Implemented Preliminary Version supporting DID
- Verifier: **Pending**
- Holder – Issuer Communication tested in Eclipse Tractus-X
- Holder – Issuer Integration in Eclipse Tractus-X: **Pending**



<https://hubs.la/Q041jyZ50>

<https://forms.office.com/e/LkYbasfXJA>



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Iñaki Paz

ipaz@lksnext.com

Innovation Project Manager
DataSpace Lead

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Thanks!

